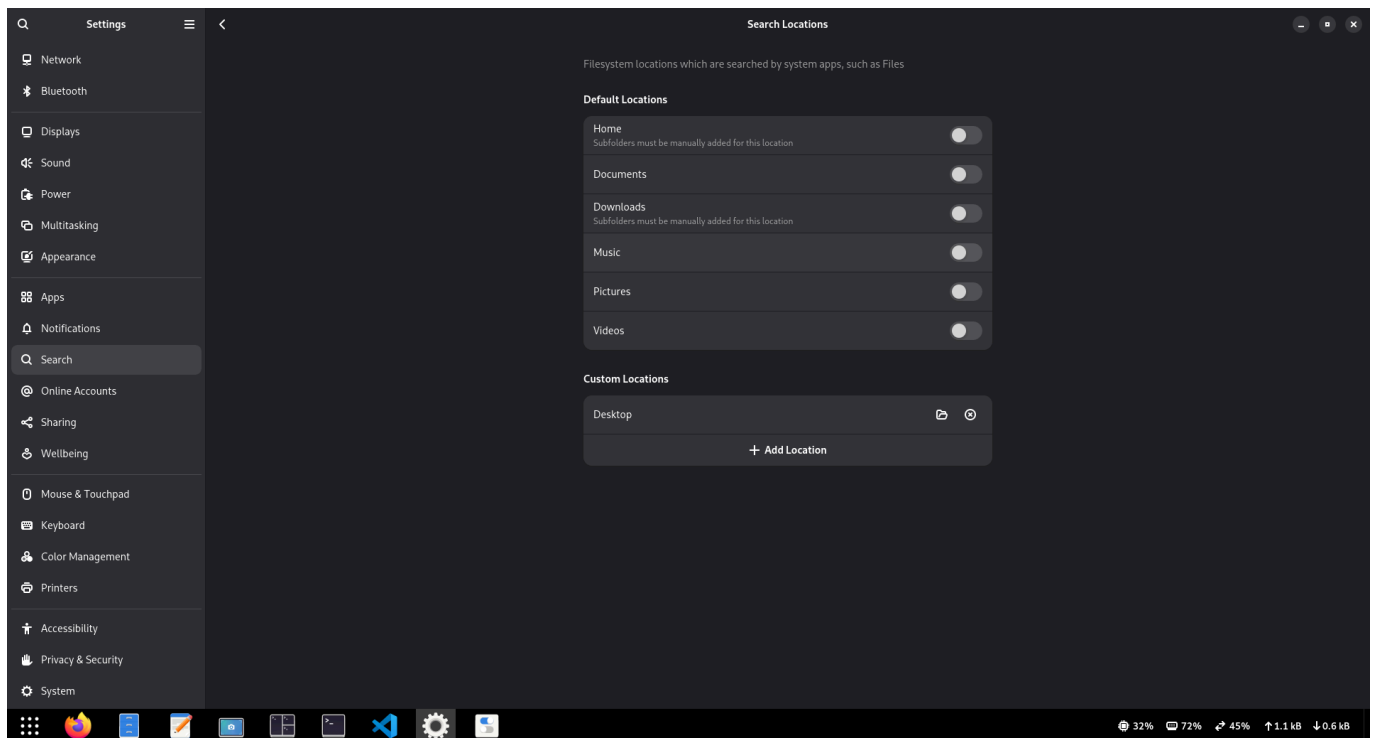
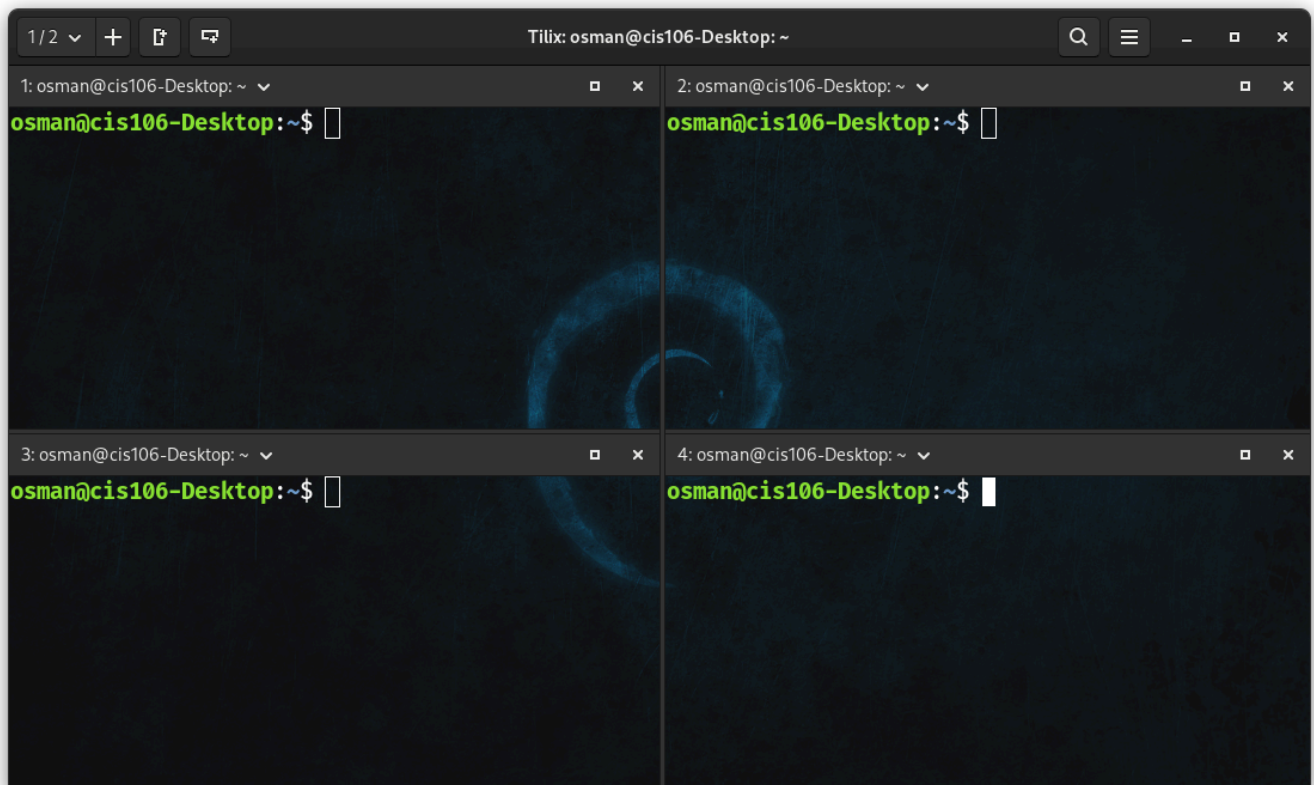


Lab 3 Submission

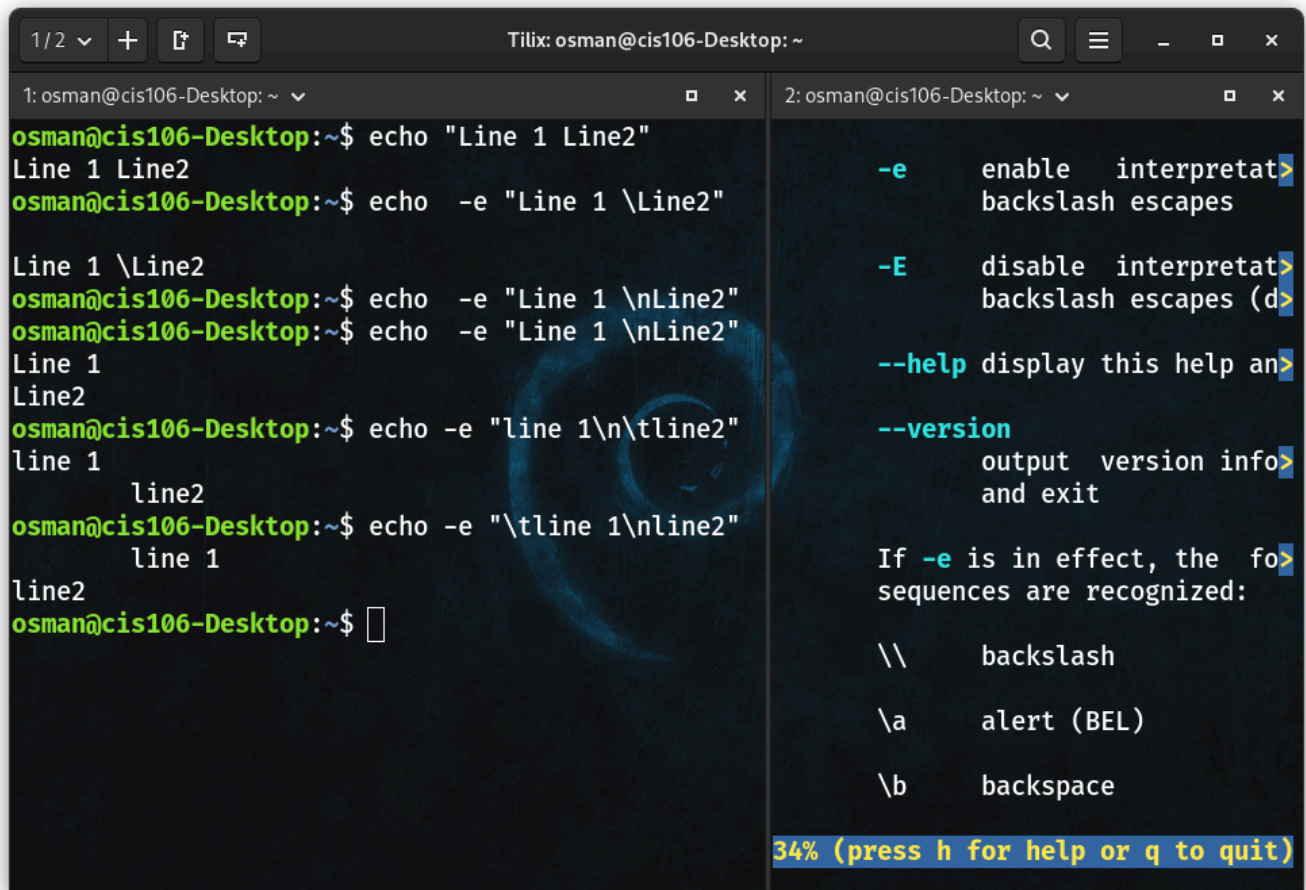
Question 1



Question 2



Question 3



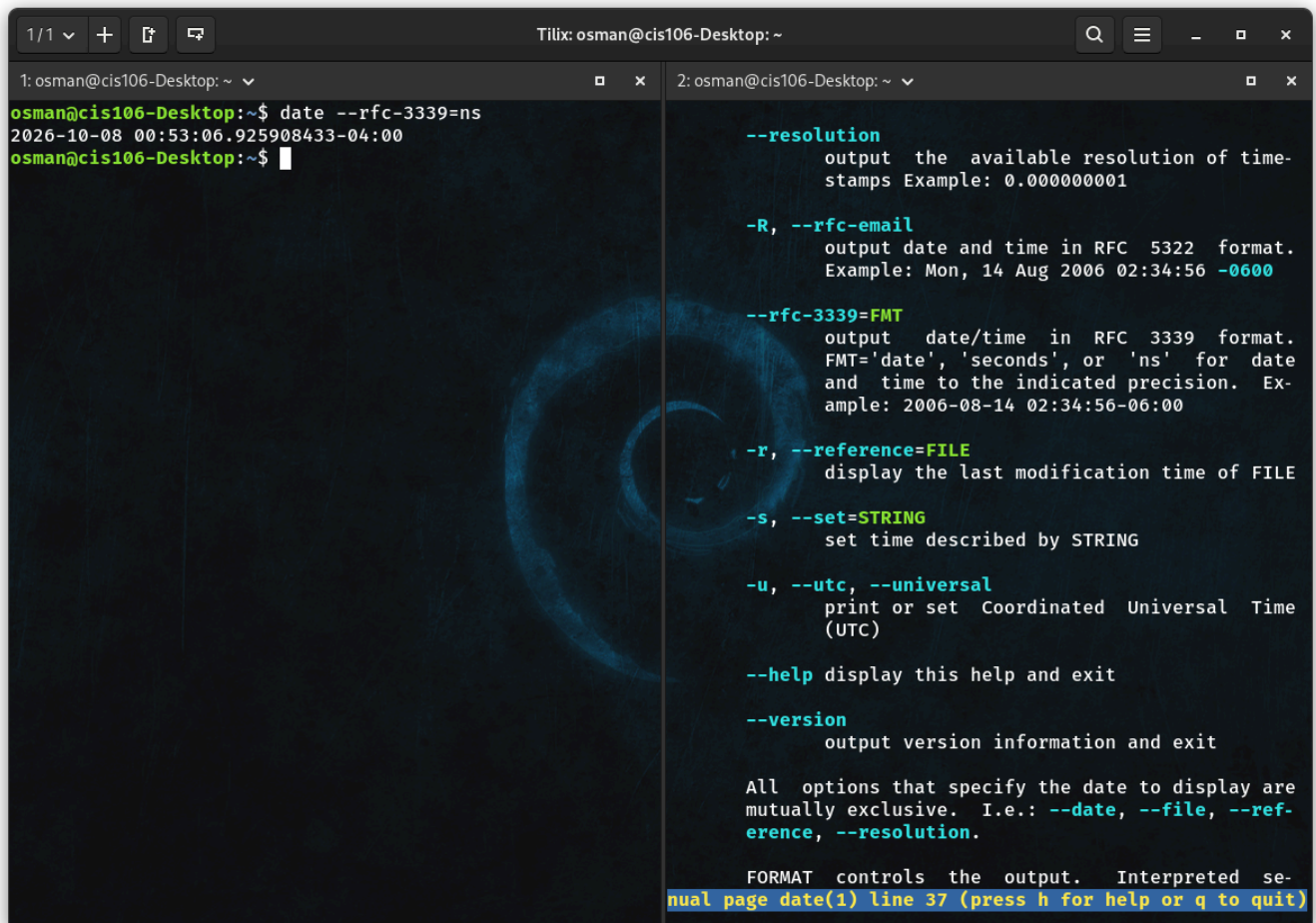
```
1/2 + [ ] [ ]
Tilix: osman@cis106-Desktop: ~
1: osman@cis106-Desktop: ~
osman@cis106-Desktop:~$ echo "Line 1 Line2"
Line 1 Line2
osman@cis106-Desktop:~$ echo -e "Line 1 \Line2"
Line 1 \Line2
osman@cis106-Desktop:~$ echo -e "Line 1 \nLine2"
osman@cis106-Desktop:~$ echo -e "Line 1 \nLine2"
Line 1
Line2
osman@cis106-Desktop:~$ echo -e "line 1\n\tline2"
line 1
    line2
osman@cis106-Desktop:~$ echo -e "\tline 1\nline2"
    line 1
line2
osman@cis106-Desktop:~$ [ ]

2: osman@cis106-Desktop: ~
-e      enable  interpretat>
        backslash escapes
-E      disable interpretat>
        backslash escapes (d>
--help  display this help an>
--version
        output version info>
        and exit
If -e is in effect, the fo>
sequences are recognized:
\\      backslash
\a      alert (BEL)
\b      backspace
34% (press h for help or q to quit)
```

Challenge Question

```
Tilix: osman@cis106-Desktop: ~  
1: osman@cis106-Desktop: ~  
osman@cis106-Desktop:~$ free -h  
              total        used        free      shared  buff/cache   available  
Mem:           3.8Gi        2.7Gi        461Mi       103Mi       1.1Gi       1.2Gi  
Swap:          2.0Gi         56Mi        2.5Gi  
osman@cis106-Desktop:~$ free -hw  
              total        used        free      shared    buffers     cache   available  
Mem:           3.8Gi        2.7Gi        459Mi       122Mi       16Mi       1.1Gi       1.2Gi  
Swap:          2.0Gi         56Mi        2.5Gi  
osman@cis106-Desktop:~$  
2: osman@cis106-Desktop: ~  
-h, --human  
    Show all output fields auto-  
    matically scaled to shortest  
    three digit unit and display  
    the units of print out. Fol-  
    lowing units are used.  
  
    B = bytes  
    Ki = kibibyte  
    Mi = mebibyte  
    Gi = gibibyte  
    Ti = tebibyte  
    Pi = pebibyte  
  
    If unit is missing, and you  
    have exbibyte of RAM or swap,  
    the number is in tebibytes and  
    columns might not be aligned  
    with header.  
  
-w, --wide  
    Switch to the wide mode. The  
    wide mode produces lines  
    longer than 80 characters. In  
    this mode buffers and cache  
    are reported in two separate  
    columns.  
  
-c, --count count  
    Display the result count  
    times. Requires the -s op-  
    tion.  
  
-l, --lohi  
    Show detailed low and high  
    memory statistics.  
  
-L, --line  
    Show output on a single line,  
    often used with the -s option  
    to show memory statistics re-  
    peatedly.  
  
-s, --seconds delay  
    continuously display the re-  
    sult delay seconds apart.  
Manual page free(1) line 101/194 76% (press h for help or q to quit)
```

```
Tilix: osman@cis106-Desktop: ~  
1: osman@cis106-Desktop: ~  
osman@cis106-Desktop:~$ uname -s -r -v -o  
Linux 6.12.107+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian  
6.12.107-1 (2026-08-29) GNU/Linux  
osman@cis106-Desktop:~$  
2: osman@cis106-Desktop: ~  
DESCRIPTION  
    Print certain system information. With no OP-  
    TION, same as -s.  
  
-a, --all  
    print all information, in the following  
    order, except omit -p and -i if unknown:  
  
-s, --kernel-name  
    print the kernel name  
  
-n, --nodename  
    print the network node hostname  
  
-r, --kernel-release  
    print the kernel release  
  
-v, --kernel-version  
    print the kernel version  
  
-m, --machine  
    print the machine hardware name  
  
-p, --processor  
    print the processor type (non-portable)  
  
-i, --hardware-platform  
    print the hardware platform (non-portable)  
  
-o, --operating-system  
    print the operating system  
  
--help display this help and exit  
  
--version  
Manual page uname(1) line 9 (press h for help or q to quit)
```



```
1: osman@cis106-Desktop: ~  
osman@cis106-Desktop:~$ date --rfc-3339=ns  
2026-10-08 00:53:06.925908433-04:00  
osman@cis106-Desktop:~$  
  
2: osman@cis106-Desktop: ~  
  
--resolution  
    output the available resolution of time-  
    stamps Example: 0.000000001  
  
-R, --rfc-email  
    output date and time in RFC 5322 format.  
    Example: Mon, 14 Aug 2006 02:34:56 -0600  
  
--rfc-3339=FMT  
    output date/time in RFC 3339 format.  
    FMT='date', 'seconds', or 'ns' for date  
    and time to the indicated precision. Ex-  
    ample: 2006-08-14 02:34:56-06:00  
  
-r, --reference=FILE  
    display the last modification time of FILE  
  
-s, --set=STRING  
    set time described by STRING  
  
-u, --utc, --universal  
    print or set Coordinated Universal Time  
    (UTC)  
  
--help display this help and exit  
  
--version  
    output version information and exit  
  
All options that specify the date to display are  
mutually exclusive. I.e.: --date, --file, --ref-  
erence, --resolution.  
  
FORMAT controls the output. Interpreted se-  
nual page date(1) line 37 (press h for help or q to quit)
```